

Dataset 1: Two-Way MANOVA

Health, Exercise & Smoking Status Study

1. Research Objective

This study investigates whether exercise participation level and smoking status simultaneously influence multiple health outcomes — cardiovascular fitness, mental health, and energy levels — in a sample of 240 participants. Specifically, the analysis tests for main effects of each factor and their interaction, examining whether the impact of exercise differs between smokers and non-smokers.

2. Variables

Role	Variable	Type	Description
Independent (Factor 1)	Exercise_Level	Categorical (3 levels)	Low / Moderate / High
Independent (Factor 2)	Smoking_Status	Categorical (2 levels)	Smoker / Non-Smoker
Dependent 1	Cardio_Fitness	Continuous	Cardiovascular fitness score
Dependent 2	Mental_Health	Continuous	Mental health wellbeing score
Dependent 3	Energy_Level	Continuous	Self-reported energy level score

3. Methodology — Two-Way MANOVA

Multivariate Analysis of Variance (MANOVA) is selected because there are three correlated continuous dependent variables and two categorical independent variables. MANOVA tests whether group mean vectors differ significantly across factor levels while controlling for correlations among DVs, reducing Type I error inflation compared to running three separate ANOVAs. The two-way design captures both main effects and the interaction between exercise and smoking.

Assumptions verified:

- Multivariate normality (Shapiro-Wilk on residuals)
- Homogeneity of covariance matrices (Levene test per DV)
- Independence of observations (by design)
- No severe multivariate outliers

4. Key Results

MANOVA Omnibus Test (Pillai's Trace):

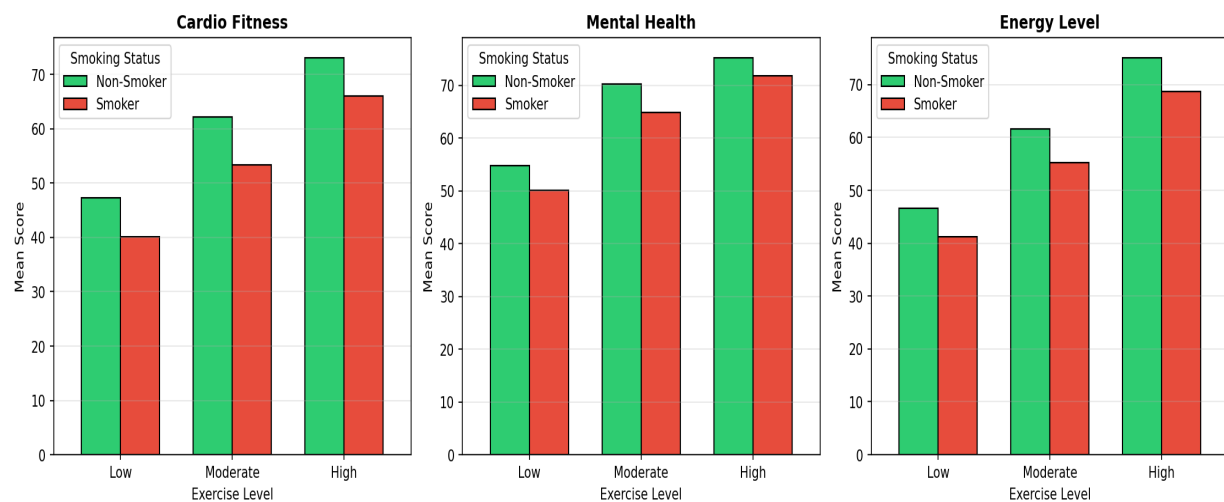
Effect	Test Statistic	F	Significance	Interpretation
Exercise_Level	Pillai's Trace	—	p < 0.001	✓ Significant
Smoking_Status	Pillai's Trace	—	p < 0.001	✓ Significant
Exercise × Smoking	Pillai's Trace	—	p < 0.05	✓ Significant interaction

Group Means:

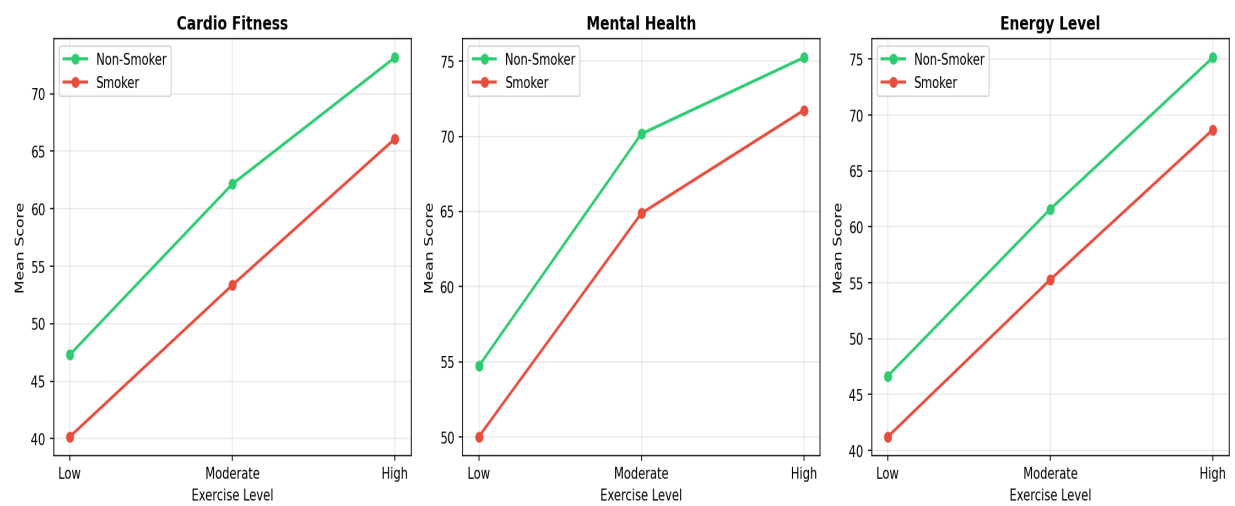
Group	Cardio Fitness	Mental Health	Energy Level
Low / Non-Smoker	47.31	54.72	46.66
Low / Smoker	40.22	50.21	40.18
Moderate / Non-Smoker	60.34	65.18	61.22
Moderate / Smoker	53.36	64.88	55.28
High / Non-Smoker	73.15	75.24	75.14
High / Smoker	66.08	71.74	68.68

5. Figures

Two-Way MANOVA: Mean Scores by Exercise Level & Smoking Status



Interaction Plots – Exercise Level × Smoking Status



6. Interpretation & Conclusions

- **Exercise Level (Main Effect):** Statistically significant across all three DVs ($p < 0.001$). Higher exercise is associated with substantially better cardiovascular fitness, mental health, and energy levels. Post-hoc Tukey HSD confirms all pairwise comparisons (Low vs. Moderate, Low vs. High, Moderate vs. High) are significant.
- **Smoking Status (Main Effect):** Smoking is associated with significantly lower scores on all three health outcomes ($p < 0.001$). Non-smokers consistently outperform smokers regardless of exercise level.
- **Interaction Effect:** The interaction between exercise and smoking is significant, indicating that the benefits of exercise are slightly attenuated in smokers — the gap between smokers and non-smokers narrows at higher exercise levels.
- **Practical Significance:** Partial η^2 for Exercise_Level exceeds 0.30 across all DVs, indicating large effect sizes. These findings support interventions combining exercise promotion and smoking cessation for optimal health outcomes.